

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (CSE) (Credit System) Degree Examinations
 May - June 2023

20CSE54 - CRYPTOGRAPHY AND NETWORK SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss any four categories of services defined by X.800.	6	L*2	1	1
b)	How Playfair cipher can be used for encryption and decryption? Discuss with suitable example.	6	L3	1	1
c)	With a neat diagram, discuss the single round of DES Algorithm. What are the strengths of DES?	8	L3	2	2
2. a)	Discuss various security mechanisms defined by X.800.	6	L2	1	1
b)	Explain different Transposition techniques with suitable example for each.	6	L3	1	1
c)	List and discuss any four types of Block Cipher modes of operations.	8	L3	2	1
3. a)	Discuss four categories of active attacks.	6	L2	1	1
b)	Differentiate between Block cipher and Stream cipher. What is the motivation for the Feistel Cipher structure?	6	L3	2	2
c)	Discuss the following concepts. i) Known plain text attacks on triple DES ii) ANSI X9.17 Pseudorandom Number Generator	8	L3	2	1
Unit – II					
4. a)	State and prove Euler's Theorem.	6	L3	3	2
b)	Discuss the following services provided by PGP. i) Authentication ii) Confidentiality	6	L2	4	1
c)	Discuss RSA Algorithm. Encrypt the message M=88 if p=17 and q=11.	8	L3	3	2
5. a)	Write Miller Rabin Algorithm for Primality testing. Discuss its working.	6	L3	3	2
b)	Discuss four applications of IPsec. List IPsec services.	6	L2	4	1
c)	List the three applications for public key cryptosystems. What are the requirements for public key cryptosystem?	8	L3	3	1
6. a)	State and prove Fermat's Theorem. List the elements in $\Phi(37)$.	6	L3	3	2
b)	Discuss the various fields of IPsec Encapsulating Security Payload (ESP) format.	6	L3	4	1
c)	Discuss Digital Signature Algorithm (DSA). Discuss two approaches to digital Signature.	8	L3	4	2
Unit – III					
7. a)	Discuss the overall operation of the Secure Socket Layer (SSL). Record protocol along with SSL record format.	10	L3	5	2
b)	Discuss the following. i) Virus structure ii) Email Viruses	10	L3	5	1
8. a)	Discuss four different phases in Handshake Protocol action.	10	L3	5	2
b)	List four general techniques that firewalls use to control access and enforce the sites security policy. Discuss three types of Firewalls with their working.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
